



4 billion rows of data!

**How Fabric Lakehouse and Power BI DirectLake help you
analyse big data**

by Bas Land





Special thanks to Fabric and Power BI Team at



This Summit presented to you by





Today

- Brief intro to Microsoft Fabric
- Introduction to Delta Lake and Lakehouse
- DirectLake Fundamentals
- Live Demo – 4 billion rows of data in Power BI
- Tips & Tricks for Optimisation
- Q&A

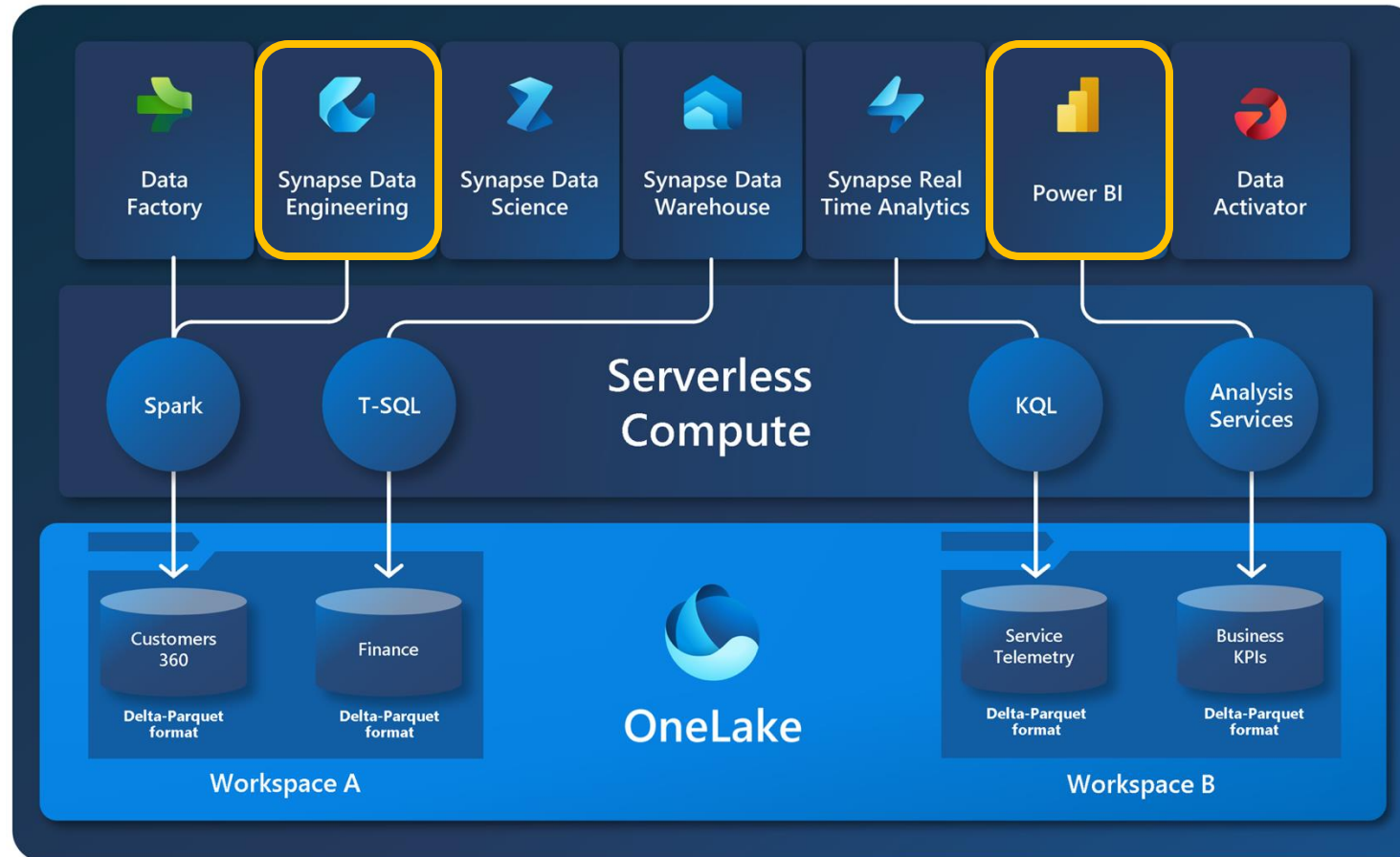




Intro to Fabric

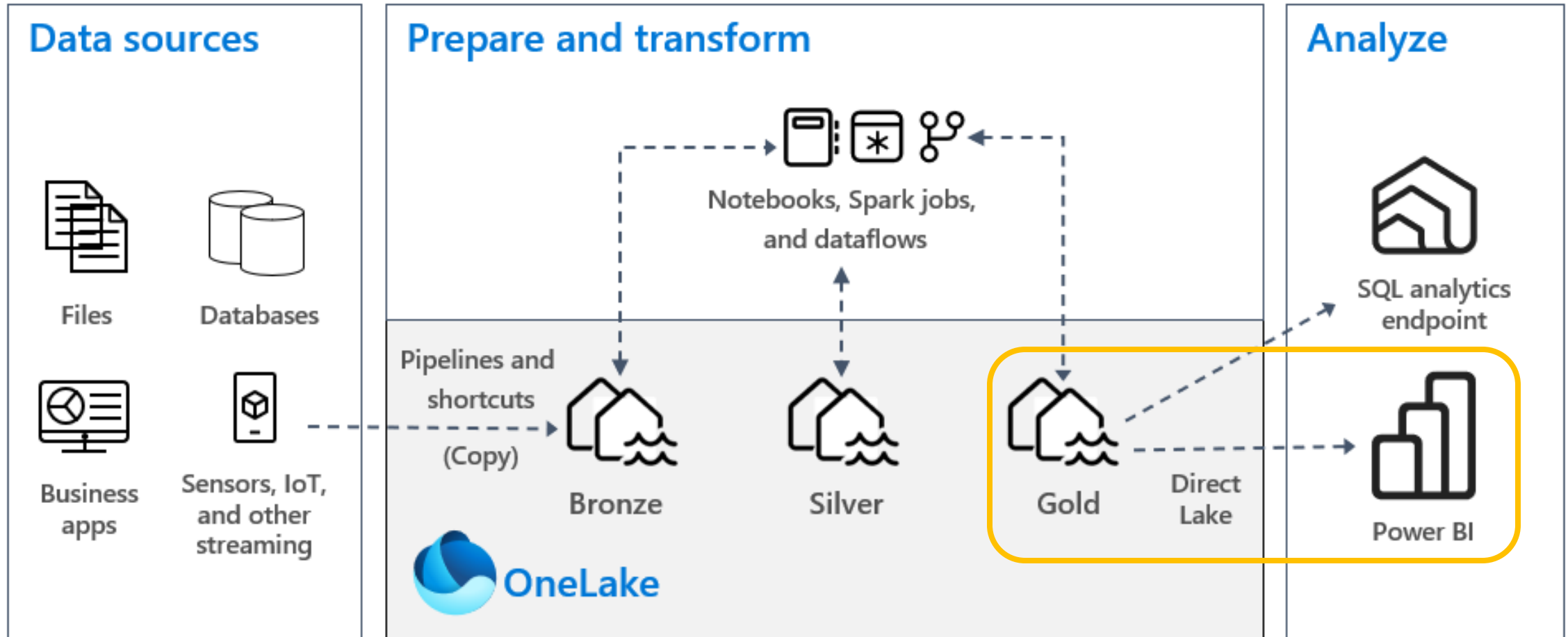


Introduction to Fabric





Introduction to Fabric





Lakehouses and Delta Lake



Lakehouse

- Not a database
- Not a data lake
- Best of both worlds!
 - Low cost
 - Queryable, structured data
- No data silos – all data pros can now work on the same physical data





Delta Lake

- Open source storage layer
- Makes 'tables' out of 'files'
- Enabling:
 - ACID transactions, Schema enforcement, Time travel
- Easily queryable, but still files on a data lake!
- We call these 'delta tables'

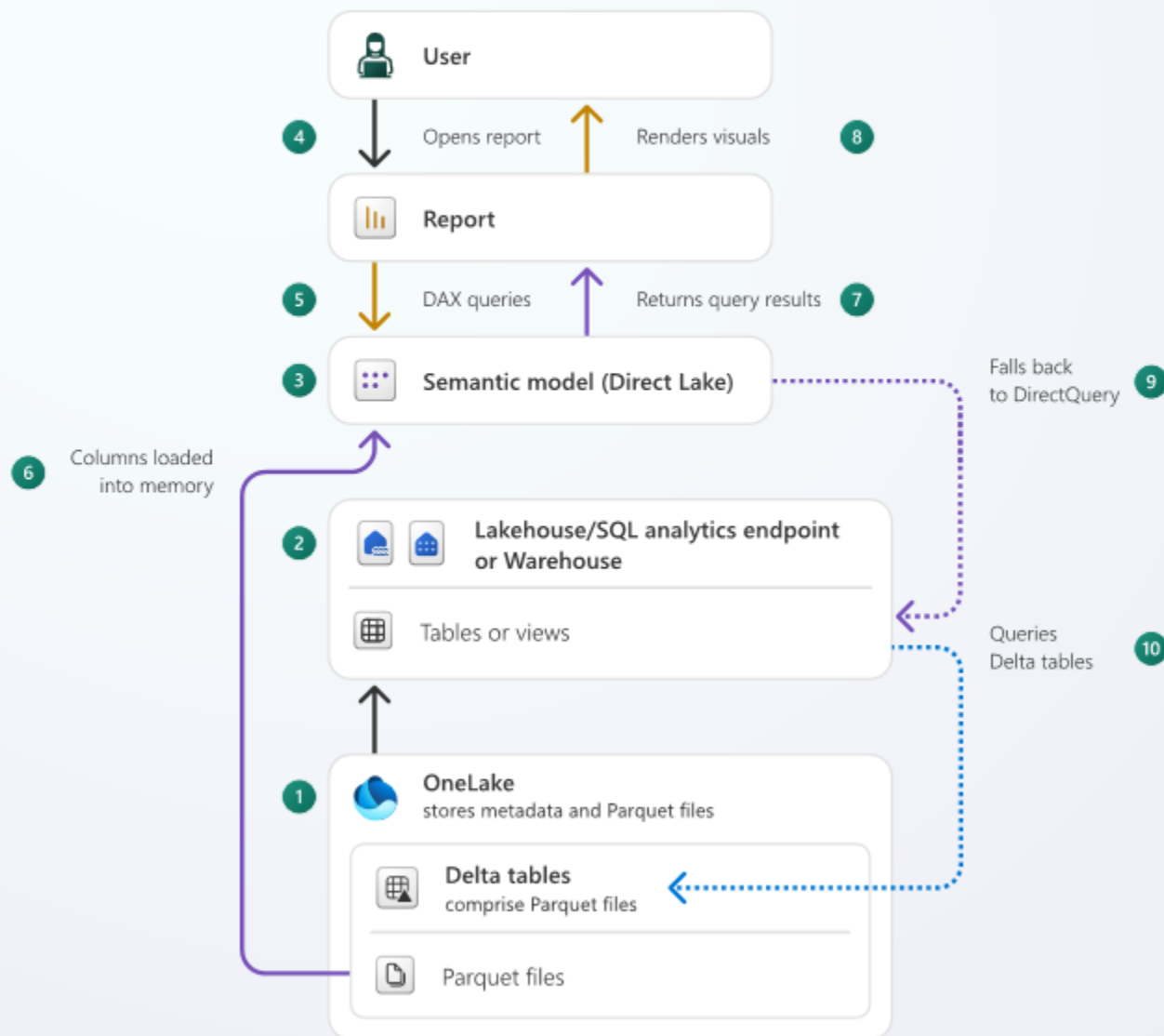




All about DirectLake



Direct Lake semantic model





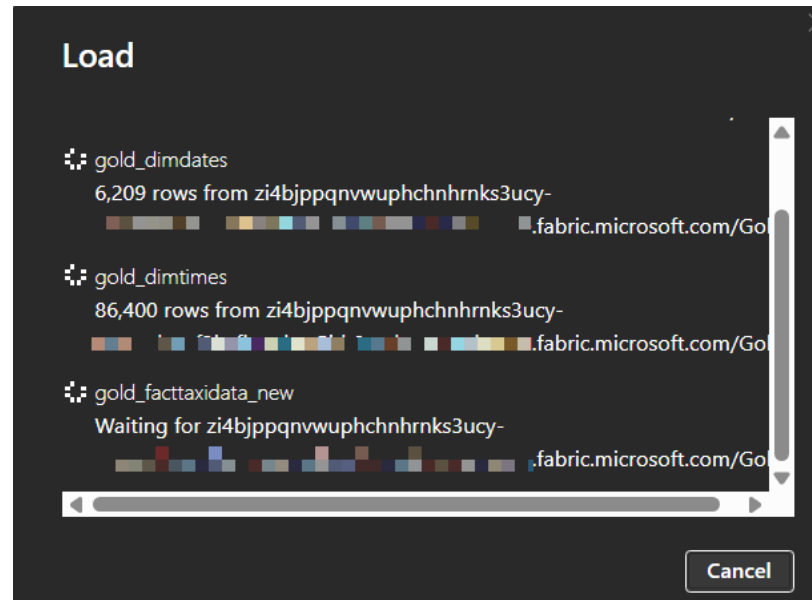
DirectLake vs Import vs DirectQuery

- Import – takes all data into Semantic Model
- DirectLake and DirectQuery keep data in the source, and take only the metadata
- DirectQuery = for databases (i.e. Azure SQL, SQL Server)
- DirectLake = for Lakehouses with Delta tables





Why not use import?





Performance considerations

- Import takes time to physically copy data to Semantic Model
- DirectQuery takes time to fetch data on user interaction
- DirectLake should bridge that gap by enabling column stored queries from Vertipaq (Semantic model / DAX engine) directly on parquet files in Delta table





Is DirectLake faster?

"IT DEPENDS"





Live Demo





Optimisation of DirectLake





Tips to optimise for DirectLake

1. Delta Lake optimisation
2. Aggregations if possible
3. Data modelling
4. Credit card 😊





Delta lake optimisation

- Use OPTIMIZE and VACUUM to keep your delta table 'clean'
- Partitioning on appropriate columns (taxi example: year, month, taxi type)
- Don't partition in too much detail!





Aggregations

- High rowcounts work, but...
- Aggregations might be better
- Look into aggregation patterns that switch the source table for a measure when selection detailed dimensions



Data modelling

- Model star schemas
- Avoid columns as much as possible – work with measures



Swipe your credit card

Fabric SKU	Parquet files per table	Row groups per table	Rows per table (millions)	Max model size on disk/OneLake (GB)	Max memory (GB) ¹
F2	1,000	1,000	300	10	3
F4	1,000	1,000	300	10	3
F8	1,000	1,000	300	10	3
F16	1,000	1,000	300	20	5
F32	1,000	1,000	300	40	10
F64/FT1/P1	5,000	5,000	1,500	Unlimited	25
F128/P2	5,000	5,000	3,000	Unlimited	50
F256/P3	5,000	5,000	6,000	Unlimited	100
F512/P4	10,000	10,000	12,000	Unlimited	200
F1024/P5	10,000	10,000	24,000	Unlimited	400
F2048	10,000	10,000	24,000	Unlimited	400





Bas Land

- BI consultant since 2013
- Co-founder of Kimura Data Intelligence:
 - Kimura Data Framework = Fabric data lake out-of-the-box
 - Strategy & Implementation consultancy on Microsoft data platform, Fabric, Power BI
- Married to Anouk, first baby on the way, we have a dachshund (😊) called Chester
- Sports: purple belt Brazilian jiu-jitsu, weight lifting, running





Q&A



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FABRIC

A scenic landscape featuring a calm lake reflecting the surrounding environment. In the foreground, a wooden house with a shingled roof sits on stilts over the water. The background is dominated by steep, rocky mountains with patches of green forest and wispy clouds. The sky is a mix of blue and soft orange, suggesting dawn or dusk. The overall mood is peaceful and majestic.

Bas Land

**4 billion rows with
DirectLake**